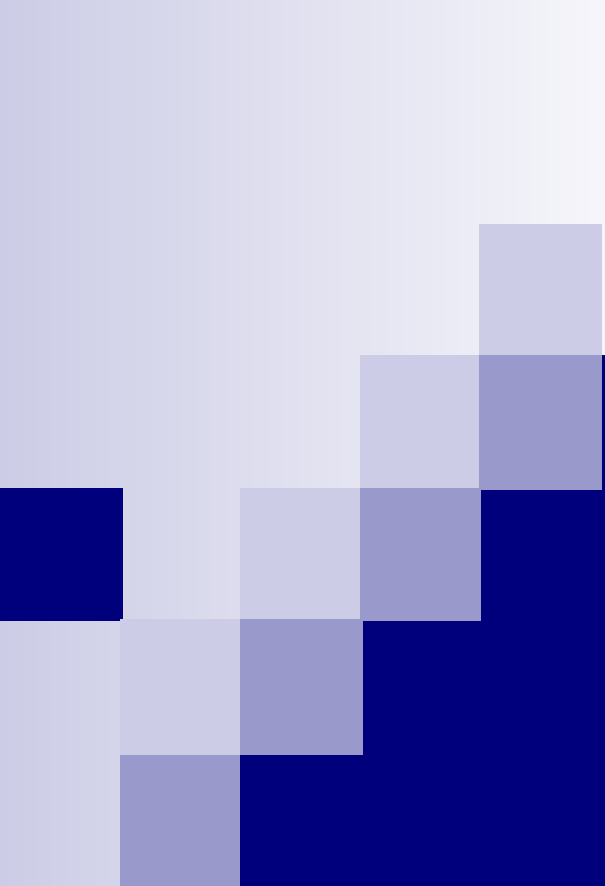




Gravitation

Newton's Law of Universal
Gravitation & Kepler's Laws



The Universal Law of Gravity

- Newton's famous apple fell on Newton's famous head, and lead to this law.
- It tells us that the force of gravity objects exert on each other depends on their masses and the distance they are separated from each other.



The Force of Gravity

- Remember $F_g = mg$?
- We've use this to approximate the force of gravity on an object near the Earth's surface.
- This formula won't work for planets and space travel.
- It won't work for objects that are far from the earth.
- For space travel, we need a better formula.

The Force of Gravity

- $F_g = Gm_1m_2/r^2$
 - F_g : Force due to gravity (N)
 - G : Universal gravitational constant
 - $6.67 \times 10^{-11} \text{ N m}^2/\text{kg}^2$
 - m_1 and m_2 : the two masses (kg)
 - r : the distance between the centers of the masses (m)
- The Universal Law of Gravity ALWAYS works, whereas $F_g = mg$ only works sometimes.

Sample Problem

mass of earth = 5.97×10^{24} kg

mass of moon = 7.36×10^{22} kg

distance between their centers =
 3.84×10^8 m

- A. How much force does the earth exert on the moon?
- B. How much force does the moon exert on the earth?

Sample Problem

radius of earth = 6.38×10^6 m

- What would be your weight if you were orbiting the Earth in a satellite at an altitude of 3,000,000 m above the earth's surface? (Note that even though you are apparently weightless, gravity is still exerting a force on your body, and this is your actual weight.)

mass of Jupiter = 1.9×10^{27} kg

distance to Jupiter = 968 million km

Sample Problem

- Sally (60 kg), an astrology buff, claims that the position of the planet Jupiter influences events in her life. She surmises this is due to its gravitational pull. Joe scoffs at Sally and says “your Labrador Retriever exerts more gravitational pull on your body than the planet Jupiter does”. Is Joe correct? (Assume a 100-lb Lab 1.0 meter away, and Jupiter at its farthest distance from Earth).



Acceleration due to gravity

- Remember $g = 9.8 \text{ m/s}^2$?
- This works fine when we are near the surface of the earth. For space travel, we need a better formula!
- What would that formula be?

Acceleration due to gravity (Gravitational field strength)

- $g = GM/r^2$
- This formula lets you calculate g anywhere if you know the distance a body is from the center of a planet.
- We can calculate the acceleration due to gravity anywhere!



Sample Problem

- What is the acceleration due to gravity at an altitude equal to the earth's radius? What about an altitude equal to twice the earth's radius?



Johannes Kepler

(1571-1630)

- Kepler developed some extremely important laws about planetary motion.
- Kepler based his laws on massive amounts of data collected by Tycho Brahe.
- Kepler's laws were used by Newton in the development of his own laws.

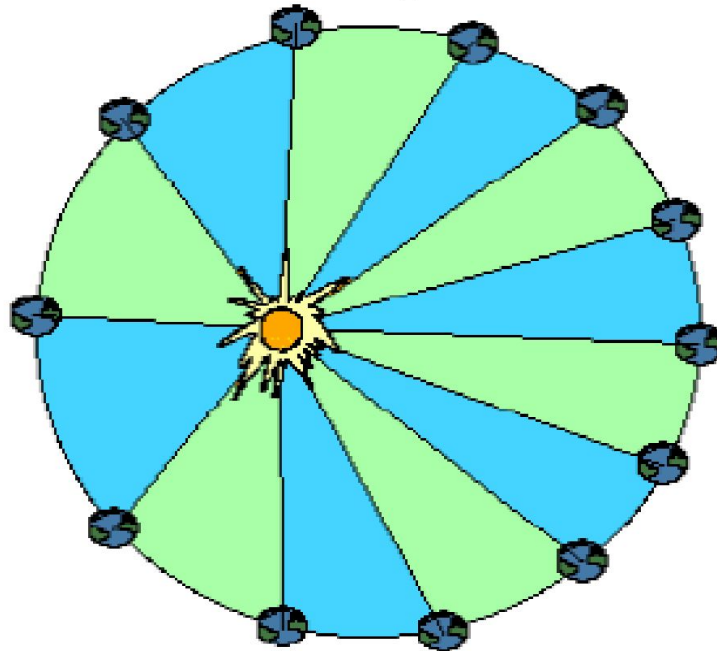


Kepler's Laws

1. Planets orbit the sun in elliptical orbits, with the sun at a focus.
2. Planets orbiting the sun carve out equal area triangles in equal times.
3. The planet's year is related to its distance from the sun in a predictable way.

Kepler's Second Law

The Law of Equal Areas

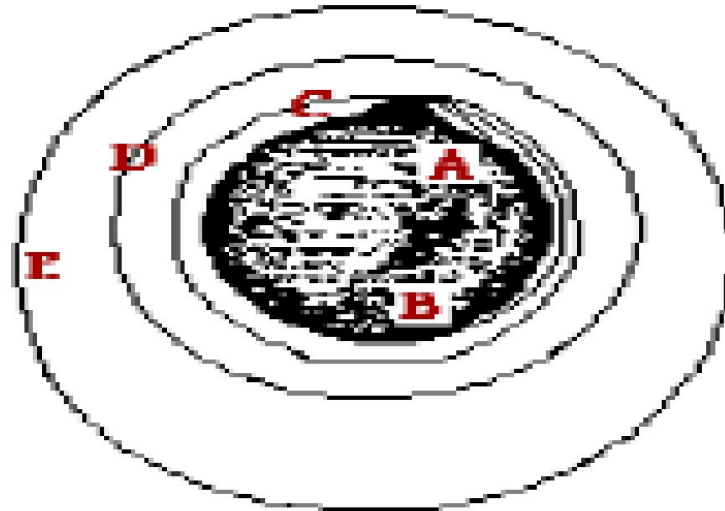


An imaginary line drawn from the sun to any planet sweeps out equal areas in equal amounts of time.



Sample Problem

- Using Newton's Law of Universal Gravitation, derive a formula to show how the period of a planet's orbit varies with the radius of that orbit. Assume a nearly circular orbit.



**A satellite is a projectile
which falls around the
earth instead of into it.**

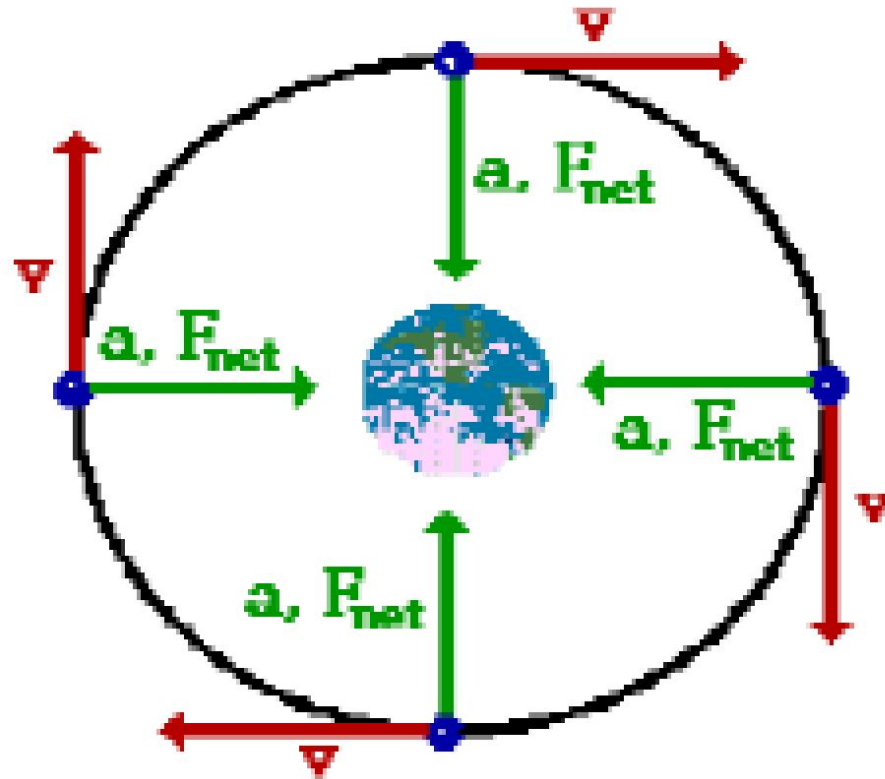
Orbital speed

- At any given altitude, there is only one speed for a stable circular orbit.
- From geometry, we can calculate what this orbital speed must be.
- At the earth's surface, if an object moves 8000 meters horizontally, the surface of the earth will drop by 5 meters vertically.
- That is how far the object will fall vertically in one second (use the 1st kinematic equation to show this).
- Therefore, an object moving at 8000 m/s will never reach the earth's surface.



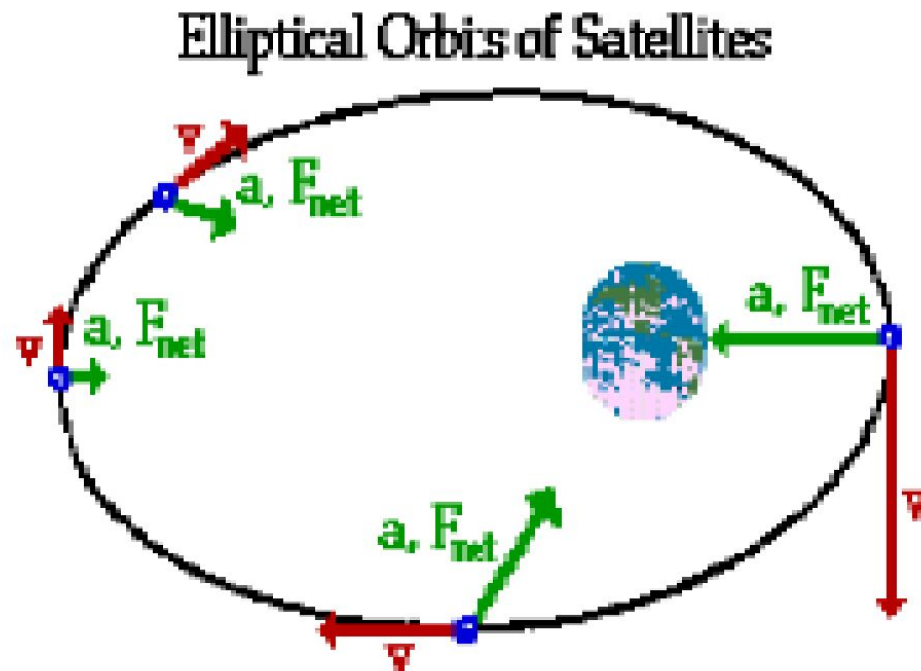
For every 8000 meters along the horizon,
the earth curves downward by 5 meters.

Some orbits are nearly circular.



Satellites encounter inward forces and accelerations and tangential velocities.

Some orbits are highly elliptical.



Even moving in elliptical motion, there is a tangential velocity and an inward acceleration and force. In this case, the a and F vectors are directed towards the central body.

Centripetal force and gravity

- The orbits we analyze mathematically will be nearly circular.
- $F_g = F_c$ (centripetal force is provided by gravity)
- $GMm/r^2 = mv^2/r$
 - The mass of the orbiting body cancels out in the expression above.
 - One of the r 's cancels as well



Sample Problem

- A. What is the velocity of a satellite in orbit about the Earth at an altitude of 25,000 km?
- B. What is the period of this satellite?



Sample Problem

- A geosynchronous satellite is one which remains above the same point on the earth. Such a satellite orbits the earth in 24 hours, thus matching the earth's rotation. How high must a geosynchronous satellite be above the surface to maintain a geosynchronous orbit?

Gravitational force

$$F = Gm_1m_2/r^2$$

F_0	G	x	m_1	x	m_2	r	r^2	x	$1/r^2$	=	FORCE FACTOR
	1	x		x				x		=	
	1	x		x				x		=	
	1	x		x				x		=	
	1	x		x				x		=	
	1	x		x				x		=	


$$F = Gm_1m_2/r^2$$

Two masses, m_1 and m_2 , are separated by a distance r between their centers exert a force, F , on each other. Determine the new force for each of the following changes.

1. m_1 is doubled
2. m_1 and m_2 are doubled
3. the separation of the masses is tripled
4. the masses are brought to $\frac{1}{4}$ of their original separation



GRAVITATIONAL ACCELERATION OR GRAVITATIONAL FIELD

$$g = GM/r^2$$

g	G	x	M	x	r	r²	x	1/r²	=	g factor
		x		x			x		=	
		x		x			x		=	
		x		x			x		=	
		x		x			x		=	
		x		x			x		=	

Mars has a mass $1/10$ that of Earth and a diameter $1/2$ that of Earth. The acceleration of a falling body near the surface of Mars is most nearly

(A) 0.25 m/s^2 (B) 0.5 m/s^2 (C) 2 m/s^2 (D) 4 m/s^2

A new planet is discovered that has twice the Earth's mass and twice the Earth's radius. On the surface of this new planet, a person who weighs 500 N on Earth would experience a gravitational force of

- (A) 125 N (B) 250 N (C) 500 N (D) 1000 N



Acceleration due to gravity

If planet X is twice the mass of earth and has an acceleration due to gravity three times that of the earth, what would its radius be?